Agentic SFT Shootout on Snowball: Teacher Traces, Methods and Parents

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Abstract

We asked which supervised fine-tuning (SFT) method improves Snowball, a 67B mixture-of-experts model, on agentic coding tasks, where the model acts over many steps in a sandbox. Five methods were compared at equal supervised tokens: plain SFT, rejection sampling across epochs (RS-multi), relay (student starts, teacher finishes), recovery (masked synthetic failures) and plateau dropping (stable-loss tokens leave the gradient). Each trained model was screened on 100 SWE-bench Verified tasks, 100 TBLite terminal tasks, GSM8K and BFCL-Parity. The main experiment used traces from GLM 5.3 at three reasoning efforts on two parents (Snowball v0 and v0.1). Every model trained on GLM traces solved fewer SWE-bench and TBLite tasks than its parent: on SWE-bench the students stopped issuing the submit command and ran into the 32k context; on TBLite they timed out. Among the methods, only relay on low-effort rows from v0.1 beat plain SFT by an interval excluding zero (SWE-bench +8 [+1, +16]), and it still trailed its parent. A lower learning rate, an earlier checkpoint and a smaller subset did not help. The same recipe on traces from two other teachers (Kimi-2.5, Qwen3.6-27B) scored at or above the parents. On the same 1,432 tasks, parent and recipe, those rows beat the GLM low-effort rows by 26 SWE-bench tasks [+17, +36] and 9 TBLite tasks [+1, +17]. The source of the traces decided the result; the methods did not. The experiment was stopped and written up; no arm was promoted.

1 Introduction

1.1 The question

Snowball is Marin's 67-billion-parameter mixture-of-experts model (2B active parameters per token). Earlier Snowball experiments compared plain SFT, rejection-sampled SFT, on-policy distillation and weight merging on non-agentic tasks; plain SFT and merging transferred skills as well as the alternatives. This experiment moves to *agentic* coding: the model runs commands in a sandbox over many steps until it has fixed a bug or finished a terminal job.

The question, as pre-registered on the tracking issue: *which SFT method, if any, improves Snowball on SWE-bench Verified and the terminal benchmarks over plain SFT at equal supervised tokens, without losing retention on GSM8K and BFCL-Parity.* Supervised tokens are the target tokens that carry the training loss; context tokens do not count.

1.2 The five methods

Plain SFT one passing teacher demonstration per task. The control.

RS-multi rejection sampling across epochs: the same tasks, but a task that recurs in a later epoch uses a different passing attempt.

Relay the student starts each task; the teacher takes over at a trigger (a format error, the context budget, or a claim that the task is done). Only the teacher's turns carry loss.

Recovery plain demonstrations in which about 20% get an inserted failure (a synthetic bad turn plus the harness's real error feedback). The inserted turns carry no loss.

Plateau dropping plain demonstrations, but tokens whose training loss has stopped changing are removed from the gradient.

Collaborators outside the team had proposed relay, recovery and plateau dropping and reported gains in their own experiments on other traces. The design regenerated all traces with one teacher, GLM 5.3, so that every method trains on the same demonstrations.

1.3 The two parents

Both parents are released Snowball versions with the same architecture, tokenizer and chat template; they are siblings from the same earlier checkpoint.

Snowball v0 a weight merge that includes an SFT on relay traces (so v0 already carries some relay training; that SFT saw 1,012 of the Kimi-2.5 trajectories used later in this report).

Snowball v0.1 a different merge followed by 14 updates of reinforcement learning. No relay training.

On our screens the two parents are not told apart: v0 minus v0.1 is +7 SWE-bench tasks [-2, +16] and -3 TBLite tasks [-12, +6].

1.4 Terms

Task family, harness SWE-smith tasks are repository bug fixes run under the *mini-swe-agent* harness; TMax tasks are terminal jobs run under *Terminus-2*. A harness is the agent program that shows the task to the model, runs its commands in a sandbox and returns the output.

Trial, episode, step a trial is one run of a harness on one task, ending with a verifier reward (1 = pass). Its sequence of model replies is an episode; one reply is a step.

Submit mini-swe-agent's command for "I am done". Episodes that never submit (for example because they hit the 32,768-token context limit) are still scored on the code they left behind.

Row, update a row is one training example built from a trace; an update is one optimizer step on 16 rows.

Plain twin the plain SFT arm on the same rows and parent as a method arm.

Composite our screening mean, (SWE-bench solved + TBLite solved) / 2 on the two 100-task sets.

1.5 Pre-registered rules

1. **Token matching.** B is the supervised-token count of the plain arm per epoch. Every arm trains on B agentic supervised tokens plus a replay slice of general SFT data of B/4, which guards retention. Plateau dropping is matched on updates instead.
2. **Learning rate.** Chosen by a four-rate sweep on pilot data (existing traces from two other teachers), then frozen. Three rates tied; the owner confirmed 1e-6 for 3 epochs.
3. **Promotion.** An arm passes when it has no decisive loss against plain on the retention screens and a decisive win on at least one agentic benchmark. Passing arms go to the full 23-benchmark tournament.
4. **Relay fallback.** If the relay pool for a family ends below its token target, a separate plain run is cut to the relay pool's size (`plain-matched-relay`).

No arm passed rule 3, and the tournament was not run.

Table 1: Rounds of the experiment, in order (dates 2026, UTC). Each round changed the training traces; the recipe (LR 1e-6, 3 epochs) stayed the same except in the LR check.

Round	Dates	Traces	Parent(s)	Runs
Pilot LR sweep	10-16/10-18	pilot (Kimi-2.5 SWE, Qwen3.6-27B TMax)	v0	4 rates
v5	10-16/10-19	GLM 5.3, max effort	v0	plain, RS-multi, recovery
v7 HIGH	10-19/10-22	GLM 5.3, high effort	v0.1 and v0	plain on both; 4 methods on v0
v7 LOW	10-20/10-23	GLM 5.3, low effort	v0.1 and v0	plain, relay; then the LOW grid
LR and dose checks	10-22/10-23	GLM 5.3, low and high	v0, v0.1	plain at 3e-7; epoch-2 checkpoint
Contributor experiment	10-22/10-23	contributor rows	v0.1 (and v0)	plain, RS-multi, recovery, plateau
Matched pair	10-23/10-24	GLM LOW and contributor, shared tasks	v0.1	2 plain runs

1.6 Rounds

2 Data

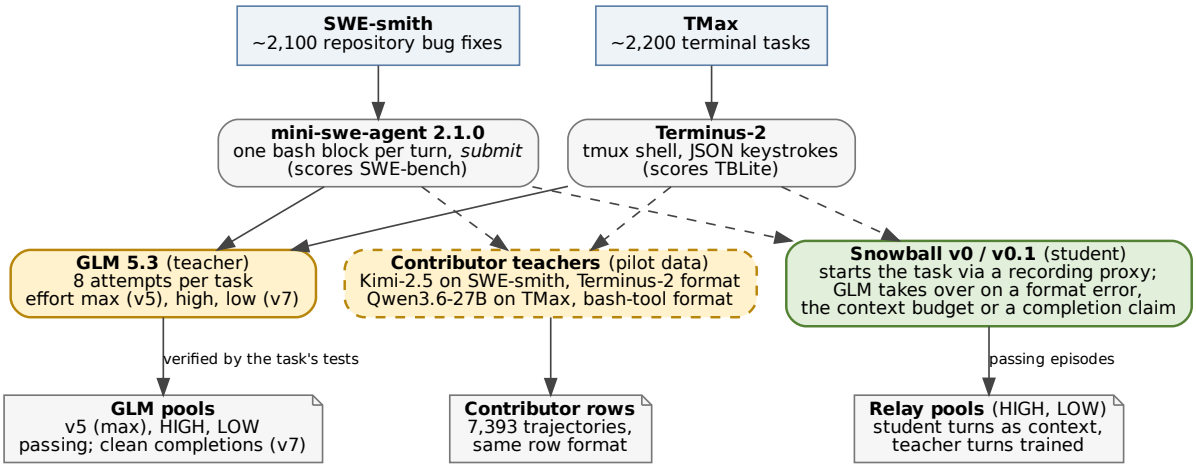


Figure 1: Step 1 as executed: one task pool, four sets of traces. Every trace is a full agent episode in a Harbor sandbox, scored pass / fail by the task's own tests. GLM 5.3 wrote three pools at different reasoning efforts; relay episodes are the student's own attempt with the teacher taking over at a trigger; the contributor traces came from the pilot data, recorded in the other harness format of each family.

2.1 Task pool

Two families, each run under the harness that the eval policy uses to score the matching benchmark (Figure 1):

TMax 2,015 terminal tasks (101 held out) from `allenai/TMax-SFT-16.5K`, run under Terminus-2.

SWE-smith 2,100 repository bug-fix tasks (105 held out) across 28 repositories, sampled from `DCAgent/swesmith-sandboxes-with_tests-oracle_verified_120s`; no SWE-bench Verified repository. Run under mini-swe-agent 2.1.0.

The pool went through five versions to remove leaks: upstream SWE-smith images kept the git history with the fix (16% of passing Kimi-2.5 pilot rows had read it); some repositories had a clean

copy of their code elsewhere in the image; 181 TMax images held unnamed verifier assets. A null agent scored 0 on 2,015 of 2,019 TMax and 2,093 of 2,100 SWE-smith tasks, and the SWE-smith gold patch scored 1 on all; the exceptions were removed. Every task was checked with 13-word n-gram matching against SWE-bench Verified, Terminal-Bench 2.1 and TBLite: 0 hits. The pool is public as `open-athena/Snowball-SFT-Shootout-Pool-2026.10`.

2.2 Teacher traces

GLM 5.3 attempted each task 8 times: SWE-smith under mini-swe-agent with a 30,000-token generation window, TMax under Terminus-2 with a 65,536-token input limit and an 1,800 s agent timeout. Every pool was scanned for leaked credentials before any row was built (0 hits).

GLM's chat template exposes three reasoning efforts. The first pool (v5) used the default, max. Its rows caused two defects in the first trained arms (Section 5.1), so a pilot measured the other settings (Table 2). The setting barely moved pass rates but changed the shape of the demonstrations several-fold. High was chosen for the main v7 pool; a plain-only pool at low effort followed. These are the HIGH and LOW pools. From v7 on, an attempt counts only if it is a *clean pass*: it ended with the agent's own finish (submit, or Terminus-2's `task_complete`).

Table 2: GLM 5.3 reasoning-effort pilot (200 tasks per family, 2 samples). Pass rates barely move; clean passes and window cut-offs on SWE-smith change several-fold.

Measure	max	high	low
TMax pass rate (%)	77.6	78.3	76.9
TMax reasoning tokens per turn, p50	709	409	81
SWE-smith pass rate (%)	61.1	68.9	65.5
SWE-smith clean pass (%)	31.1	55.6	64.2
SWE-smith trials cut by the 30k window (%)	65.6	32.8	6.6
SWE-smith completion tokens per pass, p50	8.6k	4.5k	1.4k

Relay generation. The student (a parent) runs the task through a recording proxy; GLM continues after a format error, at 75% of the context window, or after a completion claim. Relay rows keep the student's turns as context with no loss; only GLM's turns are trained. At high effort the SWE relay pools ended below their token targets (only 47-55% of GLM's passing continuations submitted, because the student's opening turns had used part of the window), which triggered the fallback of rule 4. At low effort both relay pools met their targets.

The v7 generation used about 24B GLM prompt tokens and 9,500 sandbox-hours; relay cost about as many GLM tokens as plain, plus about 2,100 H100-h of student serving.

2.3 Training rows

Every row is cut at 32,768 tokens. B is plain's agentic supervised-token count per epoch; every arm also trains on a replay slice of general SFT data (`nemotron_sft_v3`, one frozen sample) targeting B/4 (Table 3). Plain rows for v0 and v0.1 are byte-identical; only relay rows differ by parent, because the parent was the relay student. B falls from v5 to HIGH to LOW while the eligible task counts stay similar, because the episodes get shorter.

Quality of the v7 pools against v5: SWE passes ending in a submit 81.0% (HIGH) and 97.8% (LOW) against 52.3%; SWE trials cut by the window 35.1% and 7.7% against 66.0%; median TMax reasoning tokens per turn 315 and 62 against 614.

Table 3: Plain row sets by round. B is per epoch; eligible = tasks with a usable pass that fits the 32k cap. Updates = 3 epochs. Relay arms take more updates at the same B because the student's turns are context only (LOW relay v0.1: 963; HIGH relay v0: 924).

Pool	SWE eligible	TMax eligible	B SWE	B TMax	B total	Updates
v5 (max)	1,398 / 1,995	1,482 / 1,914	13.59M	15.34M	28.93M	792
v7 HIGH	1,317 / 1,995	1,531 / 1,914	7.54M	11.87M	19.42M	705
v7 LOW	1,468 / 1,995	1,578 / 1,914	4.21M	7.38M	11.60M	675
contributor	–	–	8.85M	4.05M	12.90M	678

2.4 Contributor rows

The contributor rows are the pilot sweep's traces converted to the arm row format: Kimi-2.5 on SWE-smith, recorded in Terminus-2 format, and Qwen3.6-27B on TMax with bash tool calls. Note the formats: the GLM rows use each evaluation's own harness, the contributor rows the other one. The converter reproduced the pilot's 7,393 training trajectories; 58 tasks were excluded for parity with pool v5. Of the 1,995 pilot SWE-smith tasks, 480 are in pool v5 and 1,012 were in the relay SFT merged into v0, which is why the contributor method arms ran on v0.1.

3 Methods and training recipe

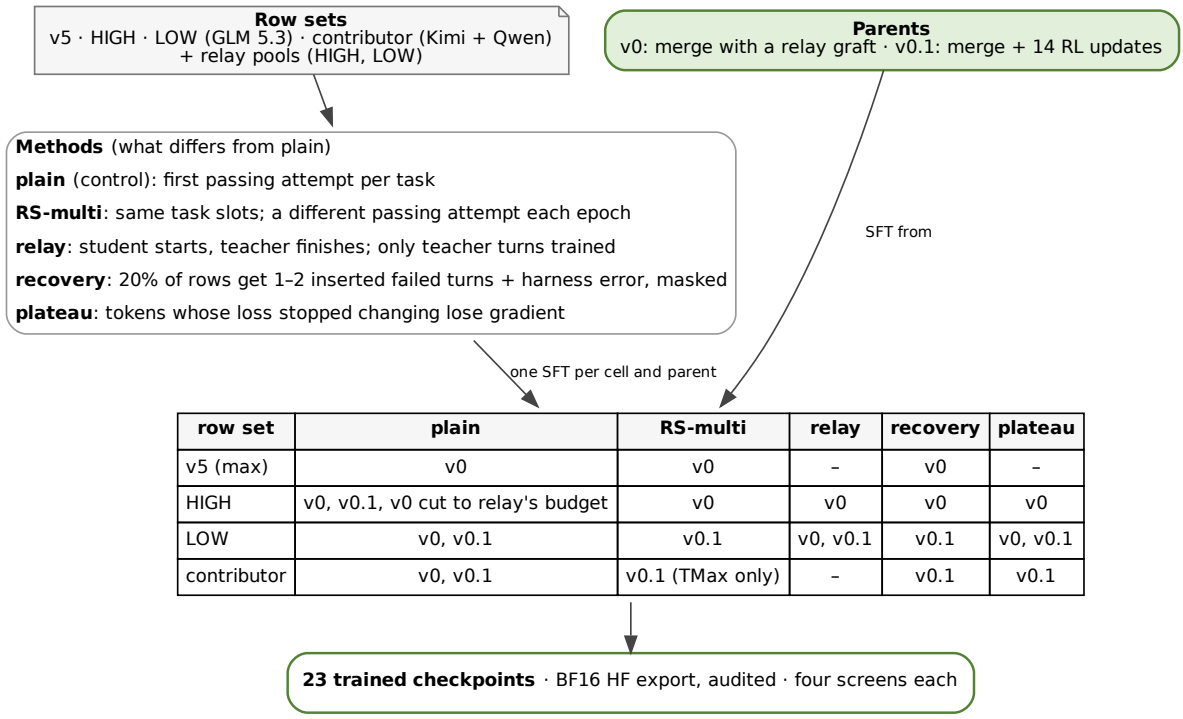


Figure 2: Step 2 as executed: five methods, four row sets, two parents, one recipe. Each cell of the grid is one SFT per listed parent; 23 arms in all. Plain is the control on every row set.

An episode becomes one tokenized row in the parent's chat template, with reasoning rendered inline. Teacher assistant turns are supervised; system, user and tool messages are context. Student prefixes in relay and injected failed replies in recovery have zero loss. Rows are not packed.

Plain the first eligible passing attempt per task, recurring across epochs in seeded permutations.

RS-multi the same task slots; a different passing attempt per epoch. Each epoch within 1.5% of plain's B. On contributor rows only TMax varies (the Kimi set has one trace per task).

Relay passing relay episodes sampled per family toward plain's budget. The student prefix changes the visited states and the loss mask, so this compares the whole relay recipe.

Recovery one or two failure points in about 20% of rows, each with one to three failed attempts, away from the first and last teacher turns, in the collection harness's own error style (Terminus-2 keeps the malformed reply and parser feedback; mini-swe-agent-style harnesses keep the feedback only).

Plateau dropping for each supervised token the losses at its previous two visits are stored; from visit three on, a token is dropped when both successive relative changes are below 0.1 (floor 0.05 nat). Dropped tokens stay in the input and return if the criterion fails. With three epochs, dropping first applies in the final epoch.

Table 4: Common arm recipe (Figure 2).

Setting	Value
Learning rate	1e-6, constant, no warmup
Epochs	3; final checkpoint, no in-loop validation
Optimizer	AdamW, weight decay 0, betas 0.9 / 0.999, clip 1.0
Global batch	16 rows; 32,768-token context per row
Objective	mean cross-entropy over active supervised tokens
Routing bias	frozen
Hardware	32 H100 GPUs (Levanter in Marin)

The final BF16 HF exports (39 shards, 502 tensors) were audited for shard presence, tensor shapes, tokenizer and chat-template identity, and the 26 routing-bias tensors against the parent.

4 Evaluation

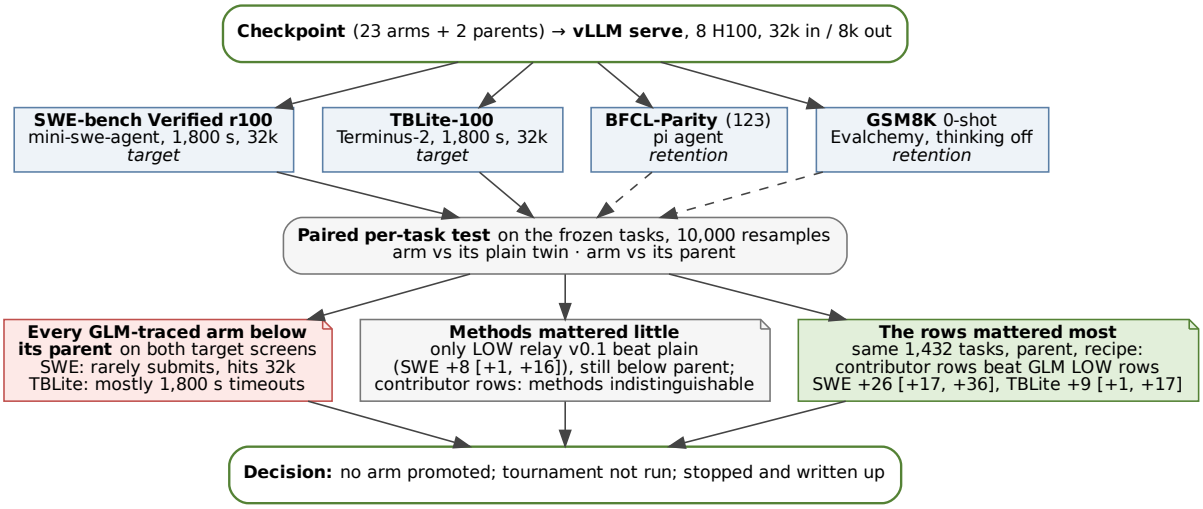


Figure 3: Step 3: four screens per checkpoint, paired per-task comparisons, and what came out.

Every checkpoint was served by vLLM on 8 H100 GPUs and scored on four screens (Table 5), one trial per task. The agentic pair decides the comparison; GSM8K and BFCL check that general skills did not break.

Table 5: The four screens. One trial per task.

Screen	Tasks	Agent / harness	Limits per trial	Metric
SWE-bench Verified r100	100	mini-swe-agent 2.1.0	1,800 s; 32,768-token context	tasks solved
TBLite-100	100	Terminus-2	1,800 s; 32,768-token context	tasks solved (reward 1)
BFCL-Parity	123	`pi` agent	1,800 s	tasks solved
GSM8K 0-shot	1,319	Evalchemy, thinking off	–	accuracy

Unscored and timeouts. A trial is unscored when no verifier result exists because of an infrastructure failure; such trials are counted both as failures and as successes, so a composite written `20.0-21.0` means two unscored trials. A TBLite trial that hits its 1,800 s limit is still verified and scored, almost always as unsolved; the tables report timeouts separately because they explain most TBLite losses. "Submits" counts SWE-bench episodes that issued the submit command; episodes that hit the context limit are still scored on their final diff.

Noise and paired comparisons. One trial per task makes every screen noisy: the only direct repeat, parent v0 on SWE-bench under two harness installs, scored 33/100 both times with 8 tasks flipping each way. Comparisons between two checkpoints therefore use a paired per-task test on the frozen tasks: the sum of per-task differences, with a 95% interval from 10,000 bootstrap resamples of tasks. We call a difference established only when the interval excludes 0. These intervals are descriptive, without multiple-comparison correction, and do not cover training-seed variation.

5 Results: the GLM experiment

Takeaway. Every student trained on GLM 5.3 traces scores below its parent on both agentic screens while GSM8K and BFCL rise. On SWE-bench the students nearly stop submitting and run into the 32k context; on TBLite most losses are timeouts. Only relay on LOW rows from v0.1 beats its plain twin by an interval excluding 0 (+8 SWE-bench tasks), and it stays below its parent.

In every table, "SWE solved (submits)" gives tasks solved and episodes ending in a submit; "TBLite solved (timeouts)" gives tasks solved out of scored and trials that hit the 1,800 s limit.

Table 6: The two parents. v0 leads on SWE-bench, v0.1 on TBLite; v0.1 times out on 49 TBLite tasks against v0's 8, mostly on tasks neither parent ever solved.

Parent	GSM8K	SWE solved (submits)	TBLite solved (timeouts)	BFCL-Parity	Composite
v0	85.2	33 (56)	20 / 99 (8)	27 / 122	26.50-27.00
v0.1	91.1	26 (36)	23 / 99 (49)	32 / 123	24.50-25.00

5.1 First round (v5): maximum-effort traces

Table 7: v5 arms: GLM 5.3 at maximum effort, parent v0, 792 updates. The arms almost never submit and time out on most TBLite tasks.

Model	GSM8K	SWE solved (submits)	TBLite solved (timeouts)	BFCL-Parity	Composite
parent v0	85.2	33 (56)	20 / 99 (8)	27 / 122	26.50-27.00
v5 plain	90.0	15 (0)	8 / 94 (67)	46 / 123	11.50-14.50
v5 RS-multi	89.7	4 (0)	3 / 90 (67)	59 / 123	3.50-8.50
v5 recovery	89.3	13 (1)	8 / 98 (63)	53 / 123	10.50-11.50

Two failures explain the loss. On SWE-bench the arms issued the submit command in 0, 0 and 1 of about 102 recorded episodes (v0: 70 of 110); every other episode ran into the 32k context, and the solves are correct diffs left in place. On TBLite 63-67 of about 95 trials hit the 1,800 s limit (v0: 8); plain takes 38 steps per trial (v0: 14) at a median of 1,075 tokens per step. Both match the training data: 47.6% of GLM's passing v5 SWE-smith trials had passed without a submit, and 72% of the trained TMax tokens were reasoning. The v5 round therefore cannot separate the methods, and the data was regenerated (v7): clean completions only, at high and low effort.

5.2 Second round (v7): HIGH and LOW rows, two parents, five methods

Table 8: v7 students of parent v0.1 (HIGH plain 705 updates, LOW plain 675, LOW relay 963).

Model	GSM8K	SWE solved (submits)	TBLite solved (timeouts)	BFCL-Parity	Composite
parent v0.1	91.1	26 (36)	23 / 99 (49)	32 / 123	24.50-25.00
HIGH plain v0.1	92.3	13 (6)	8 / 98 (41)	67 / 123	10.50-11.50
LOW plain v0.1	91.7	17 (27)	12 / 93 (42)	55 / 121	14.50-18.00
LOW relay v0.1	92.9	25 (25)	15 / 98 (64)	64 / 123	20.00-21.00

The v5 pattern repeats on clean, shorter rows. Every v7 student gains on GSM8K (+0.6 to +5.9) and BFCL (+14 to +35 tasks) and loses on both agentic screens. HIGH students submit in 1-6 of 100 SWE-bench episodes; LOW students of v0.1 still submit in 25-27.

Table 9: v7 students of parent v0 on the HIGH rows. plain-matched-relay is plain cut to the relay pool's smaller SWE budget (618 updates); relay ran 924, the others 705. No method is above HIGH plain v0 on the composite.

Model	GSM8K	SWE solved (submits)	TBLite solved (timeouts)	BFCL-Parity	Composite
parent v0	85.2	33 (56)	20 / 99 (8)	27 / 122	26.50-27.00
HIGH plain v0	87.7	17 (3)	9 / 91 (unknown)	44 / 122	13.00-17.50
plain-matched-relay v0	91.1	13 (6)	11 / 99 (70)	54 / 123	12.00-12.50
RS-multi v0	89.9	11 (6)	10 / 99 (54)	41 / 123	10.50-11.00
recovery v0	88.5	12 (4)	9 / 99 (68)	48 / 123	10.50-11.00
relay v0	89.0	13 (1)	14 / 99 (68)	48 / 123	13.50-14.00
plateau v0	89.8	14 (6)	13 / 100 (54)	49 / 123	13.50

Table 10: Paired per-task differences (A minus B), 95% task-bootstrap intervals. Bold: interval excludes 0. Of all 30 comparisons run, only one method-versus-plain row is bold; almost every student-versus-parent row is.

A	B	SWE-bench	TBLite
LOW relay v0.1	LOW plain v0.1	**+8 [+1, +16]**	+1 [-6, +7]
LOW relay v0.1	parent v0.1	-1 [-10, +8]	-7 [-18, +3]
plain-matched-relay v0	HIGH plain v0	-4 [-12, +4]	+2 [-4, +8]
RS-multi v0	HIGH plain v0	-6 [-14, +2]	+1 [-5, +8]
recovery v0	HIGH plain v0	-5 [-13, +3]	+0 [-7, +7]
relay v0	HIGH plain v0	-4 [-11, +3]	+3 [-2, +8]
plateau v0	HIGH plain v0	-3 [-10, +4]	+3 [-3, +9]
HIGH plain v0	parent v0	**-16 [-25, -7]**	**-10 [-20, -1]**
HIGH plain v0.1	parent v0.1	**-13 [-23, -4]**	**-15 [-24, -6]**
LOW plain v0.1	parent v0.1	-9 [-19, +0]	**-10 [-19, -1]**
HIGH method arms v0	parent v0	**-19 to -22**, all excluding 0	-7 to -11

5.3 What the students copy

Students reproduce the per-step style of their rows but not the episode shape (Table 11). Every student keeps the rows' tokens per step (median 112-135 against 120) and takes about twice the rows' steps (47-55 against 25). 93-97% of HIGH-row students' episodes end above 30k context and at most 6% end with a submit; the rows end with a submit in 100% of episodes at a median final context of 15.9k. The submit gap goes with the rows (LOW vs HIGH), not with the parent or the method. Context-limited trials read turn by turn show many short probe turns (one `grep` or one Python probe each) and few edits until the context runs out.

Table 11: SWE-bench episode shape: training rows against parents and selected students.

Episodes	Steps med	Tokens per step med	Final context med	Last step = submit
HIGH rows (passing only)	25	120	15,858	100%
LOW rows (passing only)	14	90	6,384	100%
pilot rows (Kimi-2.5)	14	246	12,984	100% (`task_complete`)
parent v0	37	148	29,579	56%
parent v0.1	42	75	31,724	36%
HIGH plain v0.1	50	126	32,228	6%
LOW plain v0.1	54	121	32,087	27%
LOW relay v0.1	53	127	31,975	25%
relay v0 (HIGH)	48	133	32,266	1%

Parent v0 reasons at length in nearly every step (a think block in 99.8% of steps); parent v0.1 writes no think block in 66% of its steps. Every row set is action-heavy per step, so no training data matches v0's per-step style.

5.4 Relay on LOW rows

Relay gives opposite outcomes on the two row sets. On the HIGH v0 rows (relay pool below budget) relay v0 submits in 1 of 100 episodes and is not distinguishable from plain. On the LOW v0.1 rows (token-matched) LOW relay v0.1 submits in 25 and solves 8 more SWE-bench tasks than its plain twin (+8 [+1, +16]), the only arm above its plain twin on all four screens, and still not above its parent (SWE-bench -1 [-10, +8]). Relay is a whole recipe here: masked student prefixes and more updates at the same budget (963 vs 675), so the gain cannot be attributed to one part. One run per arm.

6 Follow-up experiments

Takeaway. Within the tested ranges, the learning rate, the number of updates and the method do not account for the losses. Changing the data source does: plain SFT on the contributor traces gives students at or above their parents, and clearly above the GLM-trace students, under the same recipe and evaluation.

6.1 Learning rate and dose on GLM rows

The rate was chosen on pilot (contributor) rows: held-out loss after three epochs was 0.272 (3e-7), 0.273 (1e-6), 0.308 (3e-6) and 0.349 (1e-5); the three lower rates' screens tied within noise (composites 25.0-26.5, 31.0-32.0, 29.5-30.0) and the owner confirmed 1e-6. Two checks then asked whether the GLM students had simply been over-trained (Table 12). A lower rate was worse on every screen: 3e-7 minus 1e-6 is -7 SWE-bench tasks [-13, -1] and -6 TBLite tasks [-14, +1]. The epoch-2 checkpoint of HIGH plain v0.1 was not resolvably better (+5 [-4, +13]) and already submitted in only 3 of 100 episodes. Less exposure to the GLM rows did not restore the parent.

Table 12: Learning-rate check (plain SFT on LOW rows, parent v0) and dose check (HIGH plain v0.1 after 2 and 3 epochs).

Run	GSM8K	SWE solved (submits)	TBLite solved (timeouts)	BFCL-Parity	Composite
LOW plain v0 at 1e-6	89.30	18 (31)	13 / 97 (37)	50 / 123	15.50-17.00
LOW plain v0 at 3e-7	87.04	11 (26)	7 / 100 (40)	39 / 123	9.00
HIGH plain v0.1, epoch 2	91.43	18 (3)	9 / 98 (57)	68 / 123	13.50-14.50
HIGH plain v0.1, epoch 3	92.3	13 (6)	8 / 98 (41)	67 / 123	10.50-11.50

6.2 LOW grid

Because relay helped only on LOW rows, the methods were re-run on the LOW rows for both parents (Table 13). Paired against LOW plain of the same parent, no method's interval excludes 0 on either screen: relay v0 +6 [-2, +14] SWE; plateau v0 +7 [-1, +15]; recovery v0.1 +7 [-1, +15]; plateau v0.1 +1 [-7, +9]; RS-multi v0.1 0 [-7, +7]. LOW relay v0.1 remains the only established method gain. Every LOW arm of v0 is below its parent in point estimate on SWE-bench.

Table 13: LOW grid: all arms on the same LOW rows (11.60M supervised tokens per epoch).

Parent	Arm	GSM8K	SWE solved (submits)	TBLite solved (timeouts)	BFCL-Parity	Composite
v0	plain	89.30	18 (31)	13 / 97 (37)	50 / 123	15.50-17.00
v0	relay	88.78	24 (19)	18 / 99 (64)	50 / 123	21.00-22.00
v0	plateau	91.28	25 (31)	14 / 98 (39)	63 / 123	19.50-20.50
v0.1	plain	91.7	17 (27)	12 / 93 (42)	55 / 121	14.50-18.00
v0.1	relay	92.9	25 (25)	15 / 98 (64)	64 / 123	20.00-21.00
v0.1	plateau	91.81	18 (20)	15 / 97 (44)	74 / 123	16.50-18.00
v0.1	RS-multi	91.74	17 (20)	11 / 96 (45)	67 / 123	14.00-16.00
v0.1	recovery	93.03	24 (30)	15 / 98 (43)	58 / 122	19.50-20.50

6.3 Contributor-data experiment

The pilot student (the sweep's 1e-6 final on contributor traces) had scored above its parent in point estimate. To test the data source directly, plain SFT was re-run on the contributor rows with the arm recipe on both parents, plus RS-multi (TMax only), recovery and plateau dropping on v0.1 (Table 14).

Table 14: Contributor-data arms (Kimi-2.5 SWE-smith + Qwen3.6-27B TMax rows), 678 updates.

Arm	GSM8K	SWE solved (submits)	TBLite solved (timeouts)	BFCL-Parity	Composite
parent v0	85.2	33 (56)	20 / 99 (8)	27 / 122	26.50-27.00
contributor plain v0	90.90	35 (35)	26 / 100 (23)	68 / 123	30.50
parent v0.1	91.1	26 (36)	23 / 99 (49)	32 / 123	24.50-25.00
contributor plain v0.1	94.01	29 (32)	28 / 99 (26)	71 / 123	28.50-29.00
contributor RS-multi-TMax v0.1	94.54	29 (37)	24 / 100 (35)	69 / 123	26.50
contributor recovery v0.1	92.04	29 / 99 (29)	29 / 100 (30)	81 / 123	29.00-29.50
contributor plateau v0.1	93.9	29 (41)	30 / 100 (25)	74 / 123	29.50

All contributor students score above their parent on the composite, with every paired interval against the parent containing 0: above in point estimate, not established. Every contributor plain arm is clearly above the GLM plain arm of the same parent (+12 to +19 tasks). The contributor students keep submitting on SWE-bench (29-41 of 100 episodes; parents 36 and 56). On these rows none of RS-multi, recovery or plateau dropping differs from plain. This comparison is not matched on tokens or tasks, which motivated the matched pair.

Table 15: Contributor arms, paired differences (A minus B). Bold: interval excludes 0.

A	B	SWE-bench	TBLite
contributor plain v0	parent v0	+2 [-8, +12]	+6 [-2, +14]
contributor plain v0	HIGH plain v0	**+18 [+10, +26]**	**+16 [+9, +24]**
contributor plain v0.1	parent v0.1	+3 [-6, +12]	+4 [-6, +14]
contributor plain v0.1	HIGH plain v0.1	**+16 [+7, +25]**	**+19 [+12, +27]**
contributor plain v0.1	LOW plain v0.1	**+12 [+3, +22]**	**+13 [+5, +21]**
contributor RS-multi-TMax v0.1	contributor plain v0.1	0 [-8, +8]	-4 [-14, +6]
contributor recovery v0.1	contributor plain v0.1	+1 [-8, +10]	+1 [-8, +10]
contributor plateau v0.1	contributor plain v0.1	+0 [-7, +7]	+2 [-7, +11]

6.4 Matched pair

Two plain runs from v0.1 on exactly the same 1,432 training tasks (466 SWE-smith, 966 TMax), each keeping its whole episode per task: LOW GLM rows (4.51M agentic tokens, 306 updates) against contributor rows (5.73M, 321 updates). The pair is matched on tasks, not tokens: the contributor side has 1.27 times the supervised tokens.

Table 16: Matched pair: same 1,432 tasks, parent v0.1, same recipe.

Model	GSM8K	SWE solved (submits)	TBLite solved (timeouts)	BFCL-Parity	Composite
plain, LOW GLM shared	92.8	10 (12)	9 / 98 (39)	62 / 123	9.50-10.50
plain, contributor shared	92.9	36 (46)	18 / 100 (25)	80 / 123	27.00
parent v0.1	91.1	26 (36)	23 / 99 (49)	32 / 123	24.50-25.00

Paired: contributor shared minus LOW shared is **+26 SWE-bench tasks [+17, +36]** and **+9 TBLite tasks [+1, +17]**. The GLM side is below the parent on both screens (-16 [-24, -8]; -14 [-22, -6]); the contributor side is above the parent on SWE-bench (+10 [+1, +19]), the first contributor-versus-parent difference whose interval excludes 0, and within noise on TBLite (-5 [-15, +4]). So the drop from GLM rows and the gain from contributor rows both occur on the same tasks: the task set does not explain the gap. The GLM side used 39% of the LOW plain budget and did not land closer to the parent (SWE -7 [-14, +0] against full LOW plain), the third result in which less exposure to the GLM rows did not help. Submitting again follows the rows: 46 of 100 episodes on the contributor side, 12 on the GLM side.

6.5 BFCL-Parity by category

Every student scores far above its parent on BFCL-Parity. Most of the gain is in the 37 *irrelevance* tasks, where the model must decline because no offered tool fits: declines rise from 4/37 (v0.1) and 12/36 (v0) to 24-27/37 for every student, about equally for every training source. Correct tool calls rise from 28 (v0.1) and 15 (v0) to 30-46, with the contributor rows' edge there (46 for contributor plain v0.1 against 42 for HIGH plain v0.1). BFCL therefore works here as a check that tool calling did not break, not as a measure of the target skill.

7 Discussion

7.1 What the comparisons identify

The GLM experiment shows that the tested demonstration-training recipes can reduce agentic performance relative to the parent, and that switching the whole row set moves the scores more than any method comparison does. The row-set comparisons are bundled: teacher, collection harness, task pool, clean-completion filtering and trace shape change together between GLM and contributor data. The matched

pair removes the task set from that bundle; teacher, format and episode length remain. Budget matching counts supervised tokens, not context tokens, rows, updates or unique task states. Each arm is a single run screened on 100 tasks per screen, with descriptive intervals.

7.2 Hypotheses

None of these was tested. They are ordered by how much evidence we have, and were revised after reading a collaborating group's public account of their own SFT and RL work on a weaker Snowball checkpoint, which obtained large SFT gains from Kimi and Qwen traces and then more from reinforcement learning.[1]

1. **Teacher traces.** GLM 5.3 episodes were long, reasoning-heavy and often ended at the context limit; the students copied that. Kimi and Qwen traces did not. The matched pair is the strongest evidence we have for anything. A different teacher on the same tasks and format would test it.

2. **SFT alone does not fix action behavior.** The collaborators' best results came from SFT then RL; their trace analysis found that RL cut invalid actions from about 80% to 2% and taught the model to submit its best effort instead of running out of context. Those are exactly the failures of our GLM students. RL from a shootout checkpoint would test it.

3. **Training stack.** The collaborators report that an early relay advantage disappeared after correcting training precision and mixture-of-experts expert capacity. Our arms trained before expert capacity became configurable in our trainer; at the default capacity with expert parallelism, overflow tokens are dropped silently (found later in another experiment). Whether the shootout arms dropped tokens is not known. It can be checked by replaying one saved run.

4. **Eval protocol.** Our screens use a 32k context, 1,800 s and one trial; the collaborators use 64k, a 3x timeout and three repeats. Our GLM students produce longer episodes, so our limits may punish them more. Longer-limit checks ran on the parent only, not on a student.

5. **Parent headroom.** The collaborators' parent was a general-SFT checkpoint at 12.8% on SWE-bench; ours were merged and RL-trained parents, and v0 already carries a relay graft trained on half of the Kimi trajectories. Demonstrations may have less to add to our parents.

6. **Harness format.** Our GLM traces were recorded in each eval's own harness and hurt that eval; the contributor traces were recorded the other way round and did not. Training in the eval's own format may overwrite the parent's behavior there. But the collaborators' Kimi SWE traces are in Terminus-2 format and their SWE eval runs under Terminus-2, and that worked, so this is weaker than it first looked. Re-recording GLM traces in the other format (about 125 H100-h) would separate format from teacher.

7. **Training dose.** We trained 3 epochs. The collaborators' RL started from a 7-epoch SFT model. This would not rescue the GLM arms, which got worse, but the contributor rows were never trained past 3 epochs.

7.3 Limitations

Each agentic screen has 100 fixed tasks and one trajectory per task; the paired bootstrap resamples tasks and does not measure training-seed variation. Infrastructure retries replace only failed trials, so a record can combine trials from different serves. TBLite scores behavior under a wall-clock limit as well as a context limit. Decontamination checks cover specified fields; a zero-hit result does not certify absence of every leak. The contributor bridge on v0 is limited by v0's earlier training on half of the Kimi tasks.

[1] "Post-Training Marin's Snowball into a Coding Agent: What Worked and What Didn't", 2026-10-21, https://lukedhlee.github.io/blog/2026/10/21/turning-snowball-into-a-coding-agent/.

8 Conclusions

1. On GLM 5.3 traces, every tested arm scored below its parent on the agentic screens. Against plain SFT on the same rows, only relay on LOW rows from v0.1 differed by an interval excluding zero (SWE-bench +8 [+1, +16]), and it still scored below its parent. No arm was promoted.

2. On contributor rows, plain SFT scored above the parents in point estimate (composite 30.5 against 26.5-27.0 from v0; 28.5-29.0 against 24.5-25.0 from v0.1), with intervals containing zero; the contributor side of the matched pair cleared its parent on SWE-bench (+10 [+1, +19]). RS-multi, recovery and plateau dropping were indistinguishable from plain on these rows.

3. The rows mattered most. On the same 1,432 tasks, parent and recipe, contributor rows beat GLM LOW rows by SWE-bench +26 [+17, +36] and TBLite +9 [+1, +17]. Which property of the rows matters (teacher, harness format, episode length) is not established.

Of the design's expectations, none held: RS-multi did not beat plain on any row set; relay's reported gain appeared only on LOW rows and did not clear the parent; recovery and plateau dropping showed no resolvable gain anywhere. Three changes that reduced exposure to the GLM rows (a lower learning rate, an earlier checkpoint, a smaller subset) did not rescue the result.

Decision. The experiment was stopped and written up after the matched pair. The cross-format conversion that would separate teacher from harness format (about 125 H100-h) was not run.

Cost. About 6,070 H100-h documented: about 3,930 for training, screens and diagnostics, and about 2,140 for the parent serves that started relay episodes. GLM teacher time (about 24B prompt tokens for v7) and sandbox CPU time are not included.

Artifacts. The task pool is public (`open-athena/Snowball-SFT-Shootout-Pool-2026.10`). Six exports are on the Hugging Face Hub as private `open-athena/Snowball-67B-A2B-SFT-Shootout-*` repositories (v5 plain, RS-multi and recovery; HIGH plain v0 and v0.1; LOW plain v0.1). The reusable code (plateau dropping as a trainer option, the recording relay proxy with trace rebuild, and harness-faithful recovery rows) is in Marin pull requests; the rest of the experiment code is archived on branches of the `marin-logbooks` repository beside this report. Results by round are on the tracking issue, marin #10213.